

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	B.VANI	Reg. No.:	192472348
Section / Batch:	CSE-AI/2024	Date:	

Code of Conduct

I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.VANI

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
 - 2 questions on Discourse and Coherence
 - 2 questions on Conversational Agents and Dialog Acts
 - 2 questions on Language Generation
 - 2 questions on Machine Translation
 - 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

RUBRIC

Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:
Communication lacks clarity or confidence

Level 1:
Unable to communicate effectively

Confidence (15)

Level 4:
Displays high confidence, positive attitude and professional behavior throughout

Level 3:
Shows good confidence with minor issues

Level 2:
Low confidence and inconsistent behavior

Level 1:
Lacks confidence and professionalism

MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

Student Response:

Q2:
Question Asked:
Student Response:

Q3:
Question Asked:
Student Response:

Q4:
Question Asked:
Student Response:

Q5:
Question Asked:
Student Response:

Q6:
Question Asked:
Student Response:

Q7:
Question Asked:
Student Response:

Q8:
Question Asked:
Student Response:

Q9:
Question Asked:
Student Response:

Q10:
Question Asked:
Student Response:

RUBRIC EVALUATION

Criterion	Weightage Assigned	Level	Marks Awarded
Technical Knowledge	25	Level X	XX
Problem Solving	25	Level X	XX
Analytical Thinking	20	Level X	XX
Communication	15	Level X	XX
Confidence	15	Level X	XX

SUMMARY TABLE

Technical Knowledge (25)	Problem Solving (25)	Analytical Thinking (20)	Communication (15)	Confidence (15)	SubQ Total	Student Avg Final Mark
Level X	Level X	Level X	Level X	Level X	XX.X	XX.X

OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

DSA0306-NATURAL LANGUAGE PROCESSING

CO5 AT 3

B.VANI(192472348)

INTERVIEW QUESTIONS AND SAMPLE RESPONSES

Q1:Question Asked: What is discourse in NLP, and how does reference resolution help maintain meaning across a text?

Student Response: Discourse refers to connected sentences or utterances that together form a meaningful communication. Reference resolution identifies the entity referred to by expressions such as pronouns. For example, in "Ravi bought a laptop. He uses it every day," "He" refers to Ravi and "it" refers to the laptop. Reference resolution improves understanding by maintaining connections between entities across sentences.

Q2:

Question Asked: What is text coherence, and how can an NLP system maintain coherence in a conversation?

Student Response: Text coherence means that the ideas in a text are logically connected and easy to follow. An NLP system can maintain coherence by tracking entities, previous statements, discourse relations, and conversation context. For example, if a user says "I booked a flight to Delhi" and later says "I want to change it," the system should understand that "it" refers to the flight booking.

Q3:

Question Asked: What are dialogue acts? Explain with an example.

Student Response: A dialogue act represents the communicative intention of an utterance. Common dialogue acts include Greeting, Request, Question, Inform, Offer, and Confirmation. For example, "Can you book a flight to Delhi?" is a Request, while "The flight is tomorrow" is an Inform act. Recognizing dialogue acts helps a conversational agent choose an appropriate response.

Q4:

Question Asked: Explain the architecture and working of a conversational agent.

Student Response: A conversational agent typically consists of speech or text input processing, natural language understanding, dialogue management, response generation, and output generation. First, the system receives the user's input and identifies the intent and important entities. The dialogue manager maintains the conversation state and selects the next action. The response generator converts that action into natural language and produces the final response.

Architecture:

User Input

↓

Natural Language Understanding



Intent and Entity Extraction



Dialogue State Tracking



Dialogue Management



Response Generation



System Response

Q5:

Question Asked: What is Natural Language Generation and what are its main stages?

Student Response: Natural Language Generation, or NLG, is the process of converting structured information or semantic representations into natural-language text. Its main stages are content determination, document or discourse planning, sentence planning, lexicalization, and surface realization. The system first decides what information to communicate and then organizes and converts it into grammatically correct sentences.

Q6:

Question Asked: What is surface realization and how is it different from discourse planning?

Student Response: Surface realization is the final stage that converts a planned semantic or linguistic structure into a grammatically correct sentence. Discourse planning decides what information should be presented and in what order, while surface realization determines the actual wording, morphology, and sentence structure. For example, discourse planning may decide to communicate that a flight is delayed, while surface realization produces "Your flight has been delayed by two hours."

Q7:

Question Asked: Explain the transfer approach in machine translation.

Student Response: The transfer approach translates a source language into a target language through an intermediate source-language or language-pair-specific representation. The process generally includes source-language analysis, transfer of syntactic or semantic structures, and target-language generation. It is useful when the system needs language-specific rules between a particular source and target language.

Architecture:

Source Language

↓

Analysis

↓

Transfer

↓

Target Structure

↓

Generation

↓

Target Language

Q8:

Question Asked: What is the Interlingua approach in machine translation?

Student Response: The Interlingua approach converts the source sentence into a language-independent meaning representation. The same representation can then be converted into multiple target languages. Its major advantage is that one source language does not need a separate direct transfer system for every target language. For example, English, Tamil, and Hindi can be mapped into one common interlingual representation and then generated into the required language.

Architecture:

English → Interlingua → Tamil

Hindi → Interlingua → Tamil

English → Interlingua → Hindi

Q9:

Question Asked: What are statistical approaches in NLP?

Student Response: Statistical approaches use probabilities learned from language data to make predictions or decisions. In statistical language modeling, the probability of a word can depend on previous words, as in unigram, bigram, and trigram models. In statistical machine translation, probabilities are used to select the most likely target-language translation. These methods are useful because they can learn patterns from large corpora instead of relying only on manually written rules.

Example:

$P(\text{word}|\text{context})$

A bigram model estimates:

$$P(w_2|w_1) = \text{Count}(w_1, w_2) / \text{Count}(w_1)$$

Q10:

Question Asked: Consider a voice assistant that receives the command "Change my flight to Mumbai tomorrow with two passengers," followed by "Actually, make that Bangalore." How should the system process this conversation?

Student Response: The system should first convert the speech into text and identify the intent as changing a flight. It should extract Mumbai as the initial destination, tomorrow as the travel date, and two as the passenger count. When the user says "Actually, make that Bangalore," the system should recognize this as a correction and update only the destination from Mumbai to Bangalore while preserving the other information.

Structured state:

Intent = Change Flight

Destination = Bangalore

Travel Date = Tomorrow

Passengers = 2

The final response should confirm the updated booking information before executing the change.